Artificial Intelligence in Cars: A Detailed Study Analyzing General Public Reception

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***Abstract*—In recent times, artificial intelligence has become common in almost all sectors of engineering and technology sectors, especially in cars. AI in cars helps make them more efficient, safe, and friendly to people driving cars for the first time. This paper presents a comprehensive analysis of people's perception of some of most common AI features in cars.**

Keywords—artificial intelligence, cars, human-friendliness, trust, safety.

I. Introduction

Artificial Intelligence (AI) has become one of the essential components of the modern automobile industry [1]. With quick adoption in the automobile industry, the integration of AI technologies in cars has facilitated the development of advanced driver assistance systems (ADAS); a cluster of smart systems that work without mandated human interaction. Intelligence like handling speeds when cruise control is enabled through slope-based calculations and distance estimated auto-braking have been widely implemented in cars across various brands. Moreover, self-driving cars – a recent breakthrough – rely solely on artificial intelligence to make autonomous decisions and thus helping, revolutionize the automotive industry.

However, the use of AI in cars has also raised concerns about public reception, including the potential impact on user trust and acceptance. Especially when it comes to the decision-making phenomenon, the credibility of AI has been questioned multiple times.

Prior studies regarding the public's reception of artificial intelligence were limited. None of them included the knowledge of people on the use of artificial intelligence in cars. As covered in [2], the data collected during the study did not consider the public perception about these technologies, this missed out on a useful insight.

This paper collects research findings that provide valuable insights into the factors that affect user experiences, trust levels. It also tries to find the bias of people towards rejecting artificial intelligence in cars. The data collection has been done through a survey hosted on Google Forms [3]. This survey also wants to highlight the importance of addressing the public's perception and trust in AI-based car systems to facilitate the successful integration of these technologies into the modern automobile industry.

This paper has been divided into five sections. The first section introduces the subject going to be presented in this paper, section two looks into the details of artificial intelligence and people perceptions about it. Also, it explains the way the data captured from the participants for the survey. The next section analyses the data obtained and presents the finding. In the fourth the results were discussed and finally the paper ends with a conclusion.

II. Artificial Intelligence And Human Perception

A. Training the Artificial Intelligence Models

Like all Artificial Intelligence (AI) models, automobile AI models are trained in with real-life scenarios to enable them to make decisions and act autonomously in the real world. The training process involves feeding the AI model with vast amounts of data collected from sensors and cameras installed in the vehicle. The data is then labeled and used to train the AI model using machine learning algorithms such as supervised learning, unsupervised learning, and reinforcement learning (refer Fig. 1).

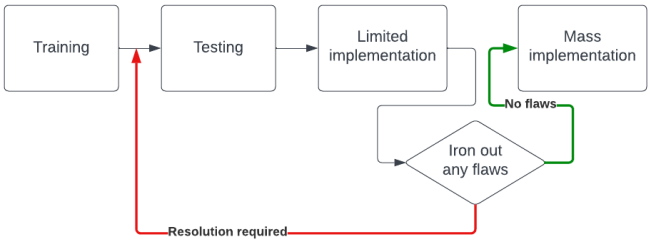


Fig 1: Simplified representation of how AI is deployed.

To maintain human friendliness, these AI models in automobiles are designed with safety and ethical considerations. The models are trained to prioritize safety over other considerations, such as convenience or speed. The models are also designed to detect and respond to unexpected events that may pose a threat to human life, such as pedestrians crossing the road.

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B. *Implementations of the Model in the Vehicle*

The first time AI was introduced in cars was in the 1980s with the development of cruise control systems. Cruise control systems [4] used a simple form of AI that could maintain a set speed without the driver's intervention. The next significant development in AI in cars occurred in the 1990s with the introduction of advanced driver assistance systems (ADAS). These systems used more sophisticated AI algorithms to enable features such as automatic braking, lane departure warnings, and adaptive cruise control [5]. In recent years, with the advancements in machine learning and deep learning techniques, AI has become more prevalent in the automotive industry. Today, AI is used in a variety of applications in cars, such as autonomous driving, predictive maintenance, and natural language processing for voice commands and infotainment systems.

C. *Capturing Perspectives from General Public*

Innovations are not good if the consumer – which most of these enhancements are aimed at – rejects them and don't want them. This is where the aforementioned human friendliness and knowledge about the existence of these features come into play. The idea behind creating this survey was to capture input from people not only in our English 202C class, but, from external sources like friends and family members also. To obtain clearer views about the general public perception, this survey was forwarded to some of my friends who lived in countries other than the United States. The survey was also designed to understand the basic knowledge of AI and their acceptance in cars: (1) Pricing; did expensiveness matter? (2) Trust in AI powered technologies such as self-driving cars (3) Knowledge about what features to be relied on AI to begin with and have they been of any use to the user – latter being an insight that hasn't been explored much.

III. Results Obtained

A. *Analzying Responder's Knowledge*

Simplicity and straightforwardness were key in designing this survey. By asking if the responder knew that there was AI being used in which was aimed to give me some insights on how knowledgeable the responder was of the topic in question. There were 32 people who responded to the survey. Most of the respondents responded that they knew what AI was and were aware of its possible uses. It can be seen in the responses that most people know that self-driving cars is a place where AI is most widely used. Others noted the added safety that comes with the implantation of AI through the use of Lane Departure Warnings and Distance Estimated Braking. Some also noted that drivers received protection from rough driving and helped avoid collisions through AI's blazing fast response times. Finally, some people highlighted the transportation benefits from the use of AI in the automobile industry by developing driver assist programs, driver risk assessments, and driver monitoring, such as monitoring a driver's eye to identify the danger of falling asleep at the wheel. Overall, the majority of the responders were aware of what the possible use cases were for AI in cars.

Parking our cars is something we do on an everyday basis. The user-interface in the reverse assisting camera is solely backed up by object recognition and distance estimation – a form of artificial intelligence. The use of AI in this particular feature however, was nowhere noted implying people responded solely based on the usefulness factor of the feature.

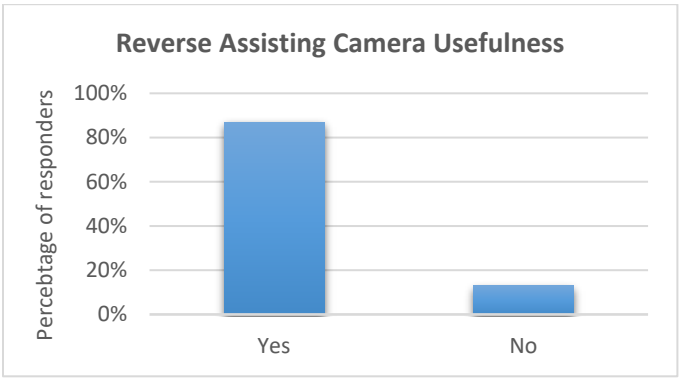


Fig 2: Summary of percentage of responders found reverse assisting camera useful.

Reverse assisting camera is one of the most successfully adopted and helpful features in modern automobiles especially with big cars where the blind spots are too large to efficiently reverse into a tight parking spot. The results obtain here show a positive correlation of the feature's usefulness. Most responders noted the beeps and sounds made when they backed their car and there was an incoming car or a wall that they were approaching. Recently, many of the cameras installed in the cars have various perspectives too going from a bottom facing angle to ultra-wide that covers a large perspective with expensive and luxury cars going as far as to cover a "360" view of the surroundings.

B. *Misconceptions about Artificial Intelligence Usages*

There are often many misconceptions about where AI is used especially when it comes to cars. Not all parts of a car's system rely on AI. Given the following options:

1. Dashboard interfaces
2. Cruise Control
3. Anti-lock Braking System (ABS)
4. Lane Keeping Assisted Driving (LKAS)
5. Distance Estimated Auto-Braking
6. Blind spot detectors
7. Tire pressure monitoring

AI is neither used for tire pressure monitoring nor anti-lock braking systems, but these options were added in between to test if people would check the right options when mingled with other more obvious choices. Many of the responders did know that tire pressure monitoring does not use AI (refer to Fig. 3) as they are the two least voted options with Lane Keeping Assisted Driving (LKAS) and Distance Estimated Auto-braking getting the highest votes. Tire pressure monitoring is done using rather trivial sensors fitting in the car's rim which just has an IC that reports back the sensor data to the car (refer to [6], Fig. 4).

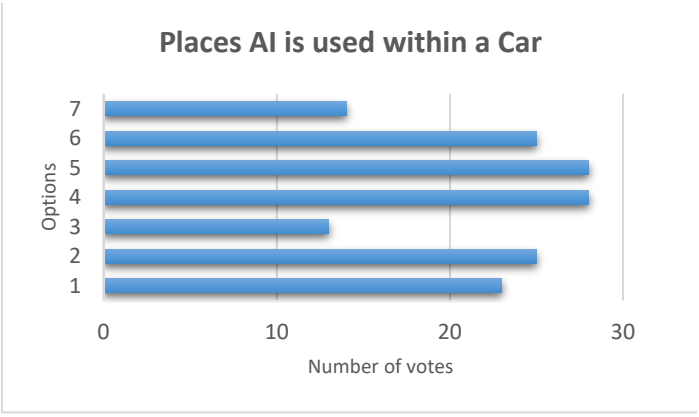


Fig 3: Votes received for features that used AI.

Similarly, Anti-lock braking system or ABS, does not rely on any kind of artificial intelligence. Rather, it relies on traditional algorithms in the car's ECU to prevent the wheels from locking b applying different amounts of braking strengths to the car's wheels.

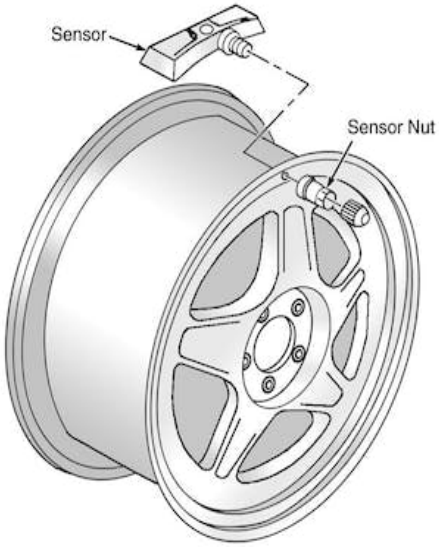


Fig 4: Location of the tire pressure sensor

C. Impact on Safety and Efficiency

Safety is one of the top problems that AI tries to solve. However, not all people feel comfortable handing off major controls.

Majority of responders were confident about having and using AI features in their cars (refer Fig. 5). It's important to keep in mind that many of the smart AI safety features have been a part of even the budget friendly cars since 2019. Most of the respondents also thought that introducing and using these AI features would significantly reduce accidents and other mishaps while driving.

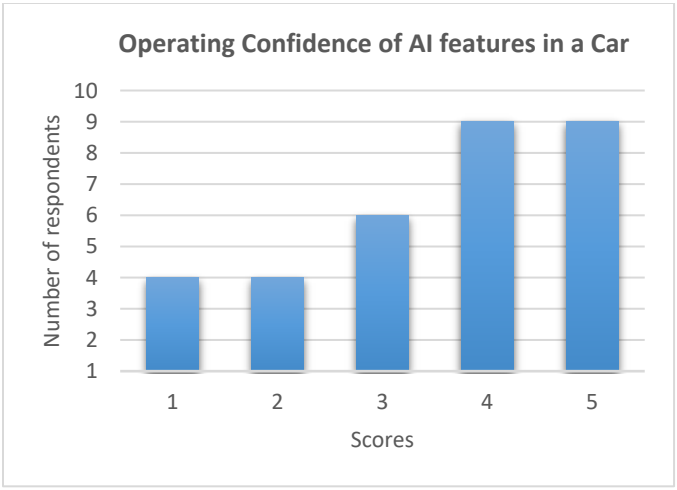


Fig 5: Aggregated scores of responders illustrating their confidence on AI features.

D. Price Factor when Getting an AI Powered Car

The success of a car depends a lot on its price. Some features more expensive than others, it adds up as more and more features are added and on the company side of things, more engineering hours are spent to ensure proper deployment is done (refer to Fig. 1). Sometimes, people prefer having a car with less features for cheap than not having a car at all. However, there are many who think that no matter the price, they will stick to their old roots and not have the new features. While some do not care about the car's price and want all the bleeding-edge things the company of their choice has to offer.

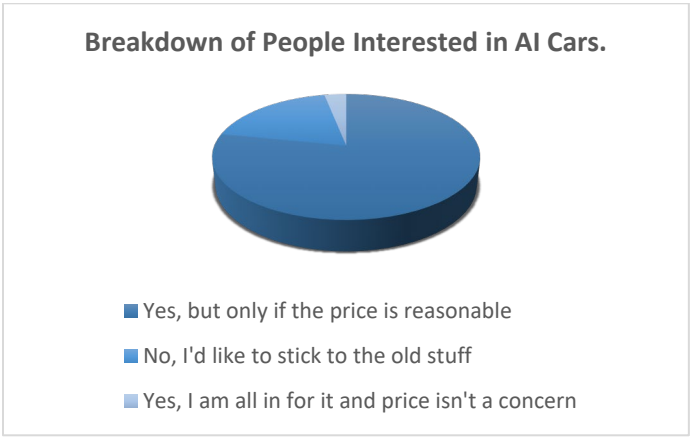


Fig 6: Distribution of people interested in getting a car with AI features.

E. Opinion Differences between Different Countries

This survey was sent out to friends and family outside of class and even the United States which ensured a wider audience was responding to the given questions. This meant that there was sometimes a stark difference in what people thought was the best use of AI and whether or they would even consider using it in the first place.

Roads and traffic in India are far different than that of the United States and United Kingdom and because of this only

15% of the respondents in India said that they would want AI in their cars. Amongst these people, one of the top concerns about self-driving cars was in the challenge of finding out a reliable way to counter all the randomness there was in the traffic – some suggesting it wasn't possible at all. Some of the responders also noted that due the presence of multiple forms of transport on the same road i.e., bicycles, motorbikes, cars, rickshaws, pedestrians on unmarked footpaths and presence of stray animals, people would take the wheel into their own hands and use their own senses to get from destination A to B.

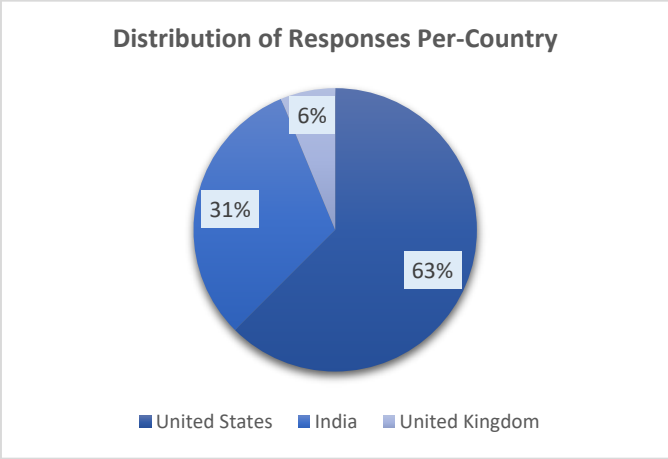


Fig 7: Visual representation of where the respondents resided.

However, the majority of respondents from the United States and all of the respondents from the United Kingdom didn't report anything similar to the respondents from India which makes sense since, roads in the US and the UK are strictly only meant for cars and sometimes bicycles and motorbikes.

However, many people weren't feeling very safe about the usage of self-driving cars on the roads.

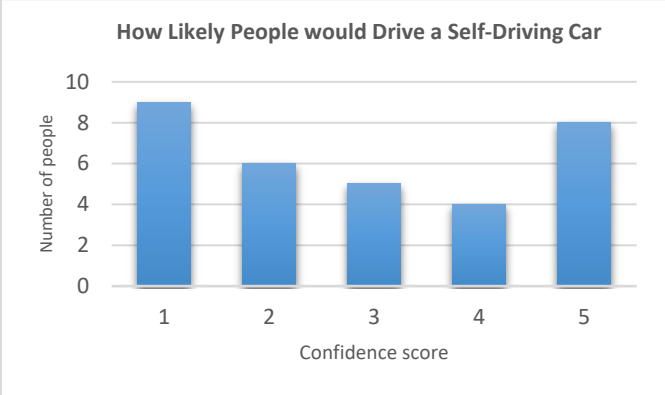


Fig 8: Likeliness of people wanting to own a self-driving car.

Self-driving cars are still very much in the early stages of full public deployment and more rigorous testing. It is also important to note that India, for example, does not have any self-driving car companies to begin with because of the aforementioned reasons about the traffic patterns and the number of things on the road.

IV. Discussion

The integration of Artificial Intelligence (AI) in cars has been a topic of discussion among technology enthusiasts and car manufacturers. The results of the survey indicate that a majority of the 32 respondents had a basic understanding of AI and its usage in cars. On average, 26 of the respondents knew about AI usage in cars. This suggests that the public is generally aware of the potential benefits of AI in cars, such as improved safety, efficiency, and convenience.

Some respondents also mentioned the use of AI-powered navigation systems, which can provide real-time traffic updates and suggest alternate routes based on traffic conditions. This not only helps drivers save time but also improves overall efficiency on the roads.

Furthermore, some respondents also mentioned the use of voice recognition and natural language processing in their cars, which are also powered by AI technology. These features enable drivers to control various functions of their car using voice commands, providing a more intuitive and convenient driving experience. It is also important to note that some respondents did not have the latest cars, and therefore, their responses were no to many questions. This could indicate a lack of exposure to AI technology in cars, which could be attributed to factors such as cost or availability. Even while keeping the aforementioned factors in mind, the survey results show that the majority of respondents who had experience with AI-powered cars were able to understand its usage and benefits. This highlights the importance of exposure and experience in increasing public awareness and understanding of AI technology.

However, this study had some limitations. First, the most important one being that some people simply did not have a car with any of the AI features. As college students tend to own older cars than the average population, some of the respondents simply were never exposed to the features in their personal cars.

Second, there's a very clear reason why the spread in Fig 7. was all over the place. For the US and the UK, Tesla's – which have a version of "Tesla Autopilot" available for beta testing still comes as a non-standard package on top of the already expensive car. As of 2023, the add-on package costs about $12,000 before taxes. This is just the price of the software as all the hardware is already present in the car. This is what call a "software lock" where a company holds certain features from being available or used until a ransom is paid.

Third, some people were simply dead against the use of AI in their cars. As one response reads, "I really don't want any technology in my car. A car should be a car, shouldn't need a computer to operate." This is where the human-friendliness factor of AI in cars comes into serious play. Some of the basic features like Lane Keeping Assistance should *just work* which would mean that there would be less fumbling around and quicker acceptance of features since there is very little interaction.

Finally, fourth, this study may not capture the full range of issues related to public perception of AI in cars. Participants may have different concerns or priorities depending on their personal experiences and backgrounds, and the study may not be able to capture these nuances.

V. Conclusion

In conclusion, as the automotive industry continues to evolve and innovate, it is important to ensure that the public is informed and educated about the advancements and benefits of AI technology in cars. This will not only improve safety and efficiency on the roads but also enhance the driving experience for consumers. Adoption is one of the key things that lead to successful technologies and AI in cars is no exception.

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